

SEID 2364

*Societal and Ethical Impact of Data Science*

# Putting It All Together

*Synthesis and Ethical Framework Construction*

Week 13

*Guiding Question — Given everything we have encountered this semester, what kind of ethical practitioner do you intend to become?*

This class serves as the principal synthesis point of the semester. Unlike previous weeks, which introduced concepts, frameworks, tools, case studies, and analytical methods, Week 13 focuses on determining which of those ideas deserve a place in each student's emerging ethical framework.

Students have spent the semester exploring ethical questions from multiple perspectives. They have considered philosophical traditions, professional standards, systems approaches, governance mechanisms, simulations, games, case studies, and personal reflection. Today is the point at which those pieces begin to come together. The emphasis throughout the class should be that frameworks are not discovered. They are constructed. Students are not searching for a correct answer. They are assembling a coherent ethical position informed by the semester's work.

## Today's Purpose

***This is not a review session.***

***This is a synthesis session.***

Our goal is to identify the ideas, tools, frameworks, and experiences that will help us construct:

**A Personal Ethical Framework**

**A Professional Ethical Framework**

The distinction between review and synthesis is important.

A review session revisits content for the purpose of recall. A synthesis session asks students to determine what remains meaningful and useful after a period of study — and then to integrate it into a coherent position they can defend.

Identification is a necessary first step: naming the ideas, principles, frameworks, and tools encountered across the semester. But identification alone is descriptive.

Synthesis requires personal judgment about which of those ideas deserve weight, and integration across sources that do not always agree (e.g., consequentialist reasoning vs. care ethics, professional codes vs. personal conviction, systems thinking vs. analytical structure).

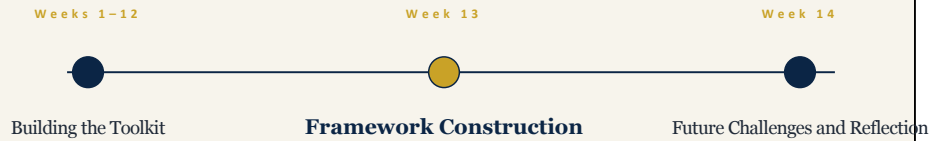
Students should understand that the final assignment is not asking them to summarize the semester, and it is not asking only for an inventory of relevant ideas. It is asking them to move from identification to synthesis — to evaluate, integrate, and commit to a Personal Ethical Framework and a Professional Ethical Framework that are recognizably their own.

As the discussion proceeds, continually return to the question: "What resources do we now possess that can help us construct these frameworks, and what judgments must we make to integrate them?"

This question becomes the organizing principle for the remainder of the class.

## Where We Are in the Course

*We are here.*



The course began with questions rather than answers. In Week 1, students encountered provocations before they possessed many analytical tools, and early responses were often intuitive, speculative, or grounded in prior experience. Weeks 1 through 12 should not be characterized as questions alone. Across that stretch students steadily accumulated a working toolkit: conceptual lenses (consequentialism, deontology, virtue ethics, care ethics, and the Big Ideas), analytical structures (EDOCA), systems frameworks (BBS), external anchors (ACM and IEEE codes, the EU AI Act), practiced moves (stakeholder analysis, harm taxonomies, bias tracing, case-study dissection, simulations and games), and reflective practice through the journal.

The arc is therefore better described as "Building the Toolkit." The original provocations remained constant; what changed was the equipment students could bring to them.

Today they revisit those questions after thirteen weeks of discussion and analysis, with a substantially expanded apparatus for answering them. This is precisely what makes synthesis meaningful — there is now a toolkit worth integrating.

Next week they will move one step further by examining future capabilities, testing the frameworks they are building, and reflecting upon the semester as a whole.

The purpose of this slide is to situate the class within the larger arc of the course and

to remind students that they enter Week 13 with more than they had in Week 1.

## Revisiting Earlier Provocations

*Take a moment to reconsider several questions encountered throughout the semester. Answer each from your current perspective before consulting your journal.*

SEID 2364

Students should answer the upcoming questions before consulting their journals. This is important.

The goal is to capture current thinking first. If students consult their journals immediately, they may simply compare themselves to their earlier answers rather than generating a fresh response.

Allow a brief period of individual reflection for each provocation. Discussion can occur later.

Explain that these questions were selected because they represent important themes that have recurred throughout the semester: Ethics, Harm, Bias, Professional Responsibility.

Together these provide a useful snapshot of how students currently understand the course.



## ***What is ethics?***

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*Answer from your current perspective.*

### **What is ethics?**

At the beginning of the semester many students approached ethics primarily as a set of rules or personal beliefs.

Over time they encountered ethical theories, professional standards, governance structures, cultural perspectives, stakeholder analysis, systems thinking, and emerging technologies.

Ask students to formulate a current answer. Avoid immediately discussing philosophical frameworks.

The purpose is to capture where they are now rather than to produce a textbook definition. The answer itself is less important than the comparison that will follow.



## ***What is harm?***

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*Answer from your current perspective.*

### **What is harm?**

This question often becomes significantly more complicated over the course of the semester. Students initially focus on direct and visible harms.

Later discussions introduce: indirect harms, systemic harms, delayed harms, stakeholder-dependent harms, unintended consequences, algorithmic harms, informational harms.

Students should consider whether their understanding of harm has expanded and whether that expansion influences how they evaluate ethical decisions.

Again, avoid discussion at this stage. The comparison comes later.

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***How can bias enter, transform, or amplify in a system?***

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*Answer from your current perspective.*

How can bias enter, transform, or amplify in a system?

Students often begin by locating bias solely within data. One of the course's recurring observations is that bias can emerge through many parts of a system: collection, selection, training, interfaces, incentives, institutions, governance, user behavior. This question serves as a reminder that ethical concerns often exist at the systems level rather than within individual components.

Students should answer from their current perspective without attempting to recall previous classroom examples.





***As a data scientist, what are your ethical obligations in system design and data management?***

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*Answer from your current perspective.*

As a data scientist, what are your ethical obligations in system design and data management?

This question intentionally places students inside the system. Rather than asking what organizations, governments, or technologies should do, it asks students to consider their own role.

Students may consider: transparency, accountability, stewardship, bias mitigation, public trust, professional responsibility, stakeholder impact.

One reason this question appears last is that it naturally points toward the final assignment. Students are no longer simply analyzing ethics. They are beginning to consider what kind of practitioner they wish to become.

## The Journal as a Historical Record

### *The Journal Was Never About Being Correct*

It was designed to:

- Record your thinking
- Preserve uncertainty
- Document change over time
- Support reflection and growth

Now students should open their journals.

The journal occupied a unique role throughout the semester. It was repeatedly described as: private, exploratory, unfinished, reflective.

Its purpose was never to generate correct answers. Instead it created a record of intellectual development.

Students should compare their current responses with their earlier entries and look for evidence of change.

Emphasize that the value of the journal lies precisely in the fact that it captured uncertainty, speculation, and developing thought.

## Looking for Change

*As you compare your responses:*

*Which assumptions remained stable?*

*Which assumptions changed?*

*Which concepts became more complicated?*

*Which questions remain unresolved?*

The goal of this slide is not evaluation. The goal is observation.

Students should identify: assumptions that remained stable, assumptions that changed, concepts that became more complicated, questions that remain unresolved.

Many students initially assume that learning should produce certainty. An important lesson from ethics is that greater understanding often produces greater awareness of complexity.

Unresolved questions are not evidence of failure. They may be evidence of intellectual growth.

## Intellectual Development

**Ethical growth does not always mean finding answers.**

*Sometimes it means recognizing complexity.*

This slide provides an opportunity to discuss what intellectual growth actually looks like.

Students frequently assume that development means replacing incorrect answers with correct answers. Ethical development often looks different.

Students may discover: increased nuance, expanded stakeholder awareness, greater appreciation for competing values, more sophisticated systems thinking, recognition of uncertainty.

This reflection serves as a bridge to the final assignment. The frameworks students construct should reflect not only what they believe but also how their thinking has evolved.

## What Are We Building?

The Final Assignment

**Construct:**

A Personal Ethical Framework

A Professional Ethical Framework

This is the pivot point of the class.

Everything before this slide focused on reflection. Everything after this slide focuses on construction.

The Personal Ethical Framework and Professional Ethical Framework are not intended as summaries of course content. They are intended as artifacts that emerge from synthesis — the student's judgment about which ideas deserve weight, integrated into a coherent position.

Identification of relevant ideas is the first step, but it is not the assignment. Students must move from inventory to integration: evaluating, weighing, reconciling tensions, and committing to a structure they can defend as their own.

Students should begin asking: What ideas matter most to me, and why? What principles do I wish to retain? What responsibilities do I recognize? What assumptions have changed? What am I choosing not to carry forward, and what does that choice say about my framework?

These questions will guide the remainder of the session.

## Framework Construction

*A framework is not selected.*

**A framework is constructed.**

A useful point to emphasize is that frameworks are assembled rather than adopted, and assembly is an act of synthesis.

Students are not being asked to choose consequentialism, deontology, virtue ethics, or care ethics and stop there. They are also not being asked simply to identify which ideas, principles, and tools they encountered.

Instead they are constructing a framework informed by multiple sources, which requires personal judgment about what matters, integration across sources that do not always agree, and the willingness to resolve or explicitly hold the tensions between them.

The arc to emphasize is identify → evaluate → integrate → commit. Identification feeds synthesis but does not substitute for it.

The remainder of the class explores resources available for that construction process. This slide serves as a conceptual transition from reflection toward synthesis.

## Resource Overview

### Resources Available for Framework Construction

|                        |                        |                          |     |
|------------------------|------------------------|--------------------------|-----|
| Reflection and Journal | Big Ideas              | Philosophical Frameworks | BBS |
| EDOCA                  | Professional Standards | Governmental Authority   |     |

The purpose of the next section is not review. Students should think of these categories as resources available for framework construction. Each resource provides a different perspective: Reflection provides personal insight. Big Ideas provide conceptual themes. Philosophical Frameworks provide ethical lenses. BBS provides systems thinking. EDOCA provides analytical structure. Professional Standards provide community expectations. Governmental Authority provides formal regulation. Students should consider which of these resources deserve inclusion in their emerging frameworks.

## Reflection and Journal

- Questions
- Assumptions
- Revisions
- Growth

The journal may ultimately prove to be one of the most important resources available to students.

Unlike theories or standards, the journal documents their own intellectual journey. Students may find: persistent values, recurring concerns, unresolved questions, significant shifts in perspective.

The journal provides evidence for framework construction because it demonstrates how thinking developed over time.

Encourage students to draw directly upon journal observations when developing their final frameworks.



## Big Ideas



The Big Ideas functioned as recurring conceptual themes throughout the semester. Unlike individual assignments, case studies, or discussions, the Big Ideas attempted to identify broader patterns that emerged repeatedly. The purpose here is to remind students that these ideas remain available as resources for framework construction. Brief expansions of each idea:

- **Ethics Requires More Than One Intelligence** — Ethics emerges where actions affect other intelligences. There is no ethics in a vacuum. Ethics arises in interaction — no interaction, no ethics. Ethics is about relationships between intelligences.
- **Ethics Requires Mediation** — Define mediation. Ethical action occurs through systems: language, law, technology, institutions, algorithms. In data science, mediation is architectural. System design, incentives, and mediation layers influence outcomes. Architecture embeds values.
- **Universal vs Cultural Morality** — Are there universal ethical principles? Are some principles absolute? Are others socially constructed? If there are universal principles, what are the axioms? Moral judgments shift across time and culture. Technological change reshapes moral boundaries. What appears obvious in one era may be contested in another.
- **Intelligent Machines Need Ethics Too** — If machines exhibit intelligence and

interact with society, they participate in ethical systems. Interaction → Consequence. Consequence → Ethical relevance. Therefore: interactive systems require ethical architecture. They classify. They allocate. They prioritize. They scale decisions. In data science, you design this architecture. As data scientists it is your responsibility to program ethical behavior.

- **Plan for Future Capabilities (Arc of Engineering)** — We are living in a period of rapid — and accelerating — technological change. Capabilities expand faster than regulatory or cultural adaptation. Technologies begin centralized. They become distributed. They become embedded and ubiquitous. When systems scale, their embedded ethical assumptions scale. Some theorists describe this acceleration as approaching a technological "singularity" — a point where capability growth outpaces human comprehension and control. Whether or not one accepts that term, we must anticipate systems whose impact exceeds present-day imagination. Therefore: we must design ethical architectures not only for current systems, but for future capabilities that do not yet fully exist. Engineering distributes judgment at scale. When systems scale, assumptions scale. Note: this idea is included here so the inventory is complete, but it will receive its full treatment in Section 6 as the bridge into Week 14. Do not dwell on it at this point.
- **Context Changes Ethical Judgment** — Even things that seem morally unambiguous may depend on context. The same behavior in different situations may be judged differently. Intent, scale, power, and consequence reshape moral evaluation. Ethics is not only about action, but about context. No definitive formulation. Trade-offs are unavoidable. Solving one dimension may worsen another.
- **Systems Have Structural Limits** — No sufficiently expressive formal system resolves every conflict. Ambiguity persists. Some questions cannot be resolved within the system that generates them — Gödel, undecidability, the halting problem. If a formal system is sufficiently expressive, there will be true statements it cannot prove. There may be ethical questions that are structurally undecidable.
- **Ethics Becomes Most Important When Problems Are Most Complex** — Ethical reasoning is easy when situations are simple. It becomes essential when trade-offs are unavoidable, stakeholders disagree, every option carries cost, and outcomes are uncertain. In complex systems, what counts as the "correct" decision may depend on how the problem is framed. Different approaches (utilitarian, rights-based, virtue-based, institutional, technical) can yield different conclusions. These are often called "wicked problems."

Ask students: Which Big Idea most influenced your thinking? Which Big Idea challenged your assumptions? Which Big Idea deserves explicit inclusion in your framework? Which ideas reinforce one another, and which create productive tensions?

## Philosophical Frameworks

### Consequentialism

*The right action is the one that produces the best outcomes.*

### Deontology

*Act according to duties and universal moral rules.*

### Virtue Ethics

*Act as a person of good character would act.*

### Care Ethics

*Respond to the needs of others within relationships.*

This slide is not intended as a review of ethical theory. Students have already encountered these traditions earlier in the semester.

The purpose here is to remind students that philosophical frameworks are resources rather than destinations.

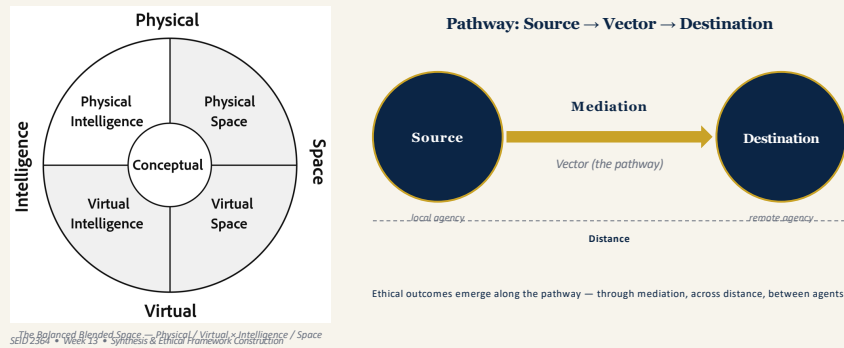
A useful observation is that ethical theories were not invented in isolation. Rather, they emerged as attempts to explain, organize, and respond to ethical challenges within society. This parallels later discussions regarding engineering, technological change, and Big Idea 5.

Students should consider: Which frameworks continue to resonate with them? Which appear incomplete? Which ideas from multiple frameworks deserve inclusion in their own ethical frameworks?

Emphasize that students are not expected to choose a single theory. Instead they may draw selectively from multiple traditions.

A framework may incorporate: consequences, duties, character, relationships, care, context — without becoming reducible to any single philosophical tradition.

## Balanced Blended Space (BBS)



18

BBS introduced a systems-oriented perspective that differs from many traditional approaches to ethical analysis.

Rather than focusing solely on individual actors, BBS encourages students to examine mediation pathways, communication processes, agency, and the movement of information through systems.

Students should reflect upon questions such as: How does mediation influence outcomes? How does distance affect responsibility? How does agency operate within complex systems? What becomes visible when we examine the pathway rather than only the endpoints?

One of the recurring lessons of the semester is that ethical outcomes often emerge from interactions among multiple participants rather than from isolated decisions.

BBS provides a way to think about those relationships.

Students do not need to adopt BBS as part of their framework. However, they should consider whether systems thinking contributes something valuable to their understanding of ethics.

## EDOCA

*A Framework for Ethical Analysis — Five Assessment Dimensions*

- **Effort / Energy** — the cost required to produce a change in system state, and who bears it.
- **Distortion** — the degree to which a signal is altered or preserved as it moves through the pathway.
- **Observability** — the degree to which an agent can perceive and interpret system state from available signals.
- **Control** — the ability of an agent to intervene in or modify system behavior at this point in the pathway.
- **Authority** — the legitimacy to assign, validate, or enforce decisions, including granting or revoking access.

EDOCA is the BBS ethical assessment lens. It provides five dimensions for analyzing what a system is actually capable of doing under real conditions of mediation, rather than what it is intended to do. The five dimensions form a causal chain: Effort enables transformation, transformation introduces Distortion, Distortion affects Observability, Observability conditions Control, and Control operates within or against Authority.

**Effort** — the cost required to produce a change in system state. Costs may be economic, time, cognitive, computational, physical, environmental, societal, or opportunity-based. Ask: what kinds of cost are present, who bears them, and how are they redistributed as distance or access changes?

**Distortion** — the degree to which a signal is altered or preserved as it moves through the pathway. Sources include noise, ambiguity, translation across domains, compression, abstraction, delayed feedback, state divergence, and many forms of bias (sampling, historical, measurement, labeling, aggregation). Distortion becomes ethically significant when degraded signals continue to guide consequential action.

**Observability** — the degree to which an agent can perceive and interpret system state from available signals. Distinguish access (can a pathway be established) from observability (can state be meaningfully understood). Observability is graded, not

binary; systems can remain accessible while becoming effectively opaque.

**Control** — the ability of an agent to intervene in or modify system behavior. Control may shift away from local agents as systems become more remote, and may revert toward local agents under degraded conditions. Control without observability is a notable source of risk.

**Authority** — the legitimacy to assign, validate, or enforce decisions, including granting, restricting, or revoking access. Authority is not the same as control. Authority often remains remote even when control shifts locally, and misalignment between authority and control is a key source of ethical risk.

Discussion prompts: Where in the pathway does each dimension sit? Are they aligned, or are they diverging? What happens when authority persists but observability or control degrade? When the dimensions diverge, the system becomes ethically unstable — that instability, not intended function, is what students should learn to evaluate.

Core principle to emphasize: **a system should be evaluated not by its intended function, but by what it is capable of doing under real conditions of mediation.**

Students should consider whether EDOCA provides a useful method for evaluating future ethical situations, and whether aspects of EDOCA belong within their Professional Ethical Framework.

## Professional Standards

- ACM Code of Ethics
- IEEE Code of Ethics

*These are exemplars, not an exhaustive list of professional codes.*

Read across them and you will see substantial alignment — and you will also see how each is flavored differently in its statements and approach.

The ACM and IEEE Codes of Ethics represent professional communities attempting to articulate expectations for responsible conduct.

Students should recognize that professional standards occupy a different role than philosophical frameworks. Philosophical frameworks help individuals reason about ethical questions. Professional standards establish expectations for members of a profession.

Discussion may include: Why do professions create ethical codes? What obligations accompany professional membership? How should professionals respond when personal beliefs conflict with professional expectations?

Students should consider whether professional standards reinforce, challenge, or extend their own ethical commitments.

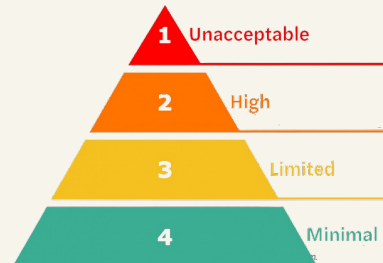
This discussion naturally prepares students to distinguish between Personal and Professional Ethical Frameworks.

## Governmental Authority



### EU AI Act — Main Principles

- Protect fundamental rights
- Ensure safety and trustworthy AI
- Risk-based regulation
- Transparency and accountability
- Harmonized single market
- Responsible innovation



*One instance of governmental authority. Data science is also shaped by privacy, sector, and anti-discrimination regimes. Practitioners move across organizations, countries, and regulatory regimes — a professional framework moves with them, but rarely unchanged. The capacity to re-analyze under changing conditions is itself a professional competency.*

SEID 2364 • Week 13 • Synthesis & Ethical Framework Construction

21

The EU AI Act represents a different form of ethical influence. Unlike professional standards, which derive authority from professional communities, government regulations derive authority from legal and political institutions. Students should consider: What role should governments play in ethical oversight? What responsibilities belong to regulators? What responsibilities remain with practitioners? What happens when regulation and professional judgment disagree? One of the recurring themes of the semester has been that ethics, governance, and technology are deeply interconnected. The EU AI Act provides a concrete example of how ethical concerns become operationalized through public policy. Students may wish to incorporate governance considerations into their Professional Ethical Frameworks.

### EU AI Act — Main Rationales

- **Protect fundamental rights** — safeguard dignity, privacy, non-discrimination, and democratic values against AI-driven harms.
- **Ensure safety and trustworthy AI** — require human oversight, robustness, accuracy, and cybersecurity, especially for high-risk systems.



- **Risk-based regulation** — tier obligations by risk (unacceptable, high, limited, minimal) and prohibit practices like social scoring and manipulative AI.
- **Transparency and accountability** — require disclosure of AI use, documentation, traceability, and clear lines of responsibility across the value chain.
- **Harmonized single market** — create uniform EU-wide rules to prevent fragmentation and provide legal certainty for developers and deployers.
- **Foster innovation responsibly** — support regulatory sandboxes and SME access while aligning AI development with European values.

AI systems are strictly prohibited if they inherently compromise human dignity, autonomy, and fundamental rights by manipulating behavior or enabling mass surveillance.

AI systems are permitted but strictly regulated if they significantly impact critical life opportunities or safety

## EU Charter of Fundamental Rights



The "fundamental rights" the EU AI Act invokes are grounded here — the Charter is the rights document the regulation operationalizes.

### Six Chapter Themes



European Union (2000). *Charter of Fundamental Rights of the European Union*. Official Journal of the European Communities, C 364/1, 18 December 2000. [https://www.europarl.europa.eu/charter/pdf/text\\_en.pdf](https://www.europarl.europa.eu/charter/pdf/text_en.pdf)

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22

The Charter of Fundamental Rights of the European Union is the constitutional rights document on which the EU AI Act stands. Proclaimed in Nice on 7 December 2000 and published in the Official Journal of the European Communities on 18 December 2000 (C 364/1), it became legally binding with the entry into force of the Treaty of Lisbon on 1 December 2009. Since then it carries the same legal value as the EU Treaties themselves.

The Charter organizes fundamental rights into six chapters: Dignity (human dignity, right to life, integrity of the person, prohibition of torture and inhuman treatment, prohibition of slavery and forced labour); Freedoms (liberty, privacy, data protection, conscience, expression, assembly, education, occupation, property, asylum); Equality (equality before the law, non-discrimination, cultural and linguistic diversity, rights of the child, the elderly, and persons with disabilities); Solidarity (workers' rights, fair working conditions, social security, health care, environmental and consumer protection); Citizens' Rights (right to vote, good administration, access to documents, freedom of movement); and Justice (effective remedy, fair trial, presumption of innocence, ne bis in idem).

The connection to the previous slide is the central pedagogical point. The EU AI Act does not invent the "fundamental rights" it claims to protect. It inherits them from

the Charter. When the Act tiers obligations by risk and prohibits certain practices outright (social scoring, manipulative AI, real-time biometric identification in public spaces), it is operationalizing Charter rights — dignity, privacy, non-discrimination, freedom of expression, effective remedy — for the AI context. The Act conforms to the Charter; the Charter does not conform to the Act.

This is a clean EDOCA Authority illustration. Authority in EDOCA is the legitimacy to assign, validate, or enforce decisions. The Act is itself an exercise of authority over providers and deployers of AI systems, but its own authority is bounded above by the Charter (constitutional ceiling) and informed laterally by deliberative work such as the High-Level Expert Group on AI's Ethics Guidelines for Trustworthy AI (April 2019) and the Assessment List for Trustworthy AI (ALTAI, 2020). The Act's seven obligations for high-risk systems map closely onto the HLEG's seven requirements — human agency and oversight, technical robustness, privacy and data governance, transparency, diversity and non-discrimination, societal and environmental wellbeing, accountability. Students should be able to trace these layers: where authority comes from, where it is exercised, and where misalignment between layers becomes ethically significant.

The framing for Week 13 synthesis is twofold. First, frameworks are layered, not flat. A professional ethical framework operates inside a stack of authorities — constitutional rights, treaties, regulation, professional codes, employer policy — each of which constrains and informs the others. Second, this stack is jurisdictional. The Charter is the EU's foundational rights document; other jurisdictions have their own (the US Bill of Rights, the Canadian Charter of Rights and Freedoms, the UK Human Rights Act, the Universal Declaration of Human Rights at the international level, and many more). The specific rights protected, the mechanisms of enforcement, and the relationship between rights and regulation all vary. A practitioner moving across jurisdictions must re-analyze which rights regime applies and how it constrains professional practice.

Discussion prompts: Which Charter rights are most directly implicated by your area of data science practice? When the EU AI Act prohibits certain AI uses outright, what Charter right is doing the prohibiting? If you were working in a jurisdiction without an equivalent constitutional rights instrument, where would the ethical authority for your professional limits come from? How does this layered structure change what you would put in your Professional Ethical Framework — and which parts of it would need to be re-analyzed if you relocated?

## From Belief to Action

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*Beliefs are important.*

**Action is where ethics becomes visible.**

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This slide serves as a transition.

Throughout the semester students have discussed ideas, principles, frameworks, and theories. The question now becomes: How do those ideas influence actual decisions? Ethics becomes visible through action. Many individuals can articulate ethical principles. The greater challenge is applying those principles consistently when decisions become difficult.

Students should begin thinking about how their frameworks will guide behavior rather than merely express beliefs.

This distinction will become important as they construct both personal and professional frameworks.

## Personal Ethical Framework

### Possible Components



Students often find it easier to discuss ethics in the abstract than to articulate their own commitments. This slide asks them to begin identifying those commitments. Possible discussion questions include: What principles matter most to you? Which values have remained stable throughout the semester? Which obligations do you recognize regardless of profession? What boundaries are important to maintain? Students should recognize that a Personal Ethical Framework may continue evolving throughout their lives. The goal is not permanence. The goal is clarity regarding current commitments and priorities.

## Professional Ethical Framework

### Possible Components



Professional frameworks introduce additional considerations. Professionals operate within systems that include: employers, clients, users, regulators, professional communities, stakeholders. Students should consider how these relationships create responsibilities that extend beyond personal belief. Discussion may include: accountability, transparency, public trust, stewardship, governance, professional competence. The purpose is not to separate personal and professional ethics completely but to recognize that professional roles introduce additional obligations.

## Relationship Between the Two

Q 1

***Must they align?***

Q 2

***Can they differ?***

Q 3

***What happens when  
they conflict?***

This slide marks the transition. Up to this point students have been constructing — selecting Big Ideas, philosophical traditions, professional standards, and governmental authorities, and weighing how they fit together into a personal and a professional ethical framework. From here forward, the work is no longer constructing the framework. It is taking a position within it. The three questions on the slide are not meant to be resolved; they are meant to expose the points where students will have to commit. Name this pivot explicitly when delivering the slide. This slide should remain exploratory. Avoid attempting to resolve the question. Instead encourage students to consider: Must personal and professional ethics always align? What happens when they conflict? How should professionals respond to disagreement? Can professional obligations ever override personal preferences? This discussion often reveals tensions that students may wish to address directly within their frameworks. The existence of tension does not indicate failure. It indicates complexity.

## Big Idea 5

### Plan for Future Capabilities

*(Arc of Engineering)*

This slide introduces the final Big Idea and serves as the culmination of the course's conceptual arc.

Big Idea 5 asks students to consider future capabilities rather than present technologies.

A recurring observation throughout the semester is that technological capability often changes more rapidly than ethical theory, governance, or social understanding. Students should begin considering whether the frameworks they are constructing can remain useful in the face of future change.

This slide intentionally shifts attention away from current examples and toward future possibilities.



## Ethical Theory and Social Change

*Music theory often follows musical practice.*

*Ethical theory often follows technological and societal change.*

The analogy to music theory is useful here. Music theory did not create music. Rather, music theory emerged as a way to describe, analyze, and understand musical practice.

Similarly, ethical theories often emerge in response to social and technological change.

Discussion may revisit examples where: technologies appeared before regulations, capabilities emerged before ethical consensus, societies adapted after innovation occurred.

The purpose of this slide is not to diminish ethical theory but to place it within a broader process of social adaptation.

Students should consider how future capabilities may challenge existing frameworks.

## What Must Remain Stable?

***Which principles should remain constant?***

***Which principles may need to adapt?***

This is a stress-testing exercise.

Students should examine their emerging frameworks and identify: principles that appear foundational, principles that may require adaptation, assumptions that depend upon current technologies, commitments that should survive technological change.

Discussion may include: privacy, autonomy, fairness, transparency, responsibility, human dignity.

The objective is not agreement but reflection. Students should begin distinguishing between enduring principles and context-dependent practices.



***How should an ethical framework  
respond to capabilities that do not yet  
exist?***

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The purpose of this question is to remain unresolved. Students should not attempt to answer it definitively.

Instead, they should consider: How should an ethical framework respond to capabilities that do not yet exist?

This question encapsulates many themes from the semester: uncertainty, complexity, governance, systems thinking, future technologies, responsibility.

It also provides a bridge into Week 14, where students will continue examining future challenges and emerging capabilities.

## Final Assignment

Construct:

Personal Ethical Framework

Professional Ethical Framework

Use evidence from:

Journal

Big Ideas

Ethical Frameworks

BBS

EDOCA

Professional Standards

EU AI Act

Course Discussions

This slide returns students to the practical objective of the class.

The assignment is not a separate activity from the semester. It is a synthesis of the semester.

Encourage students to draw upon: journal entries, provocations, Big Ideas, philosophical frameworks, BBS, EDOCA, professional standards, governance discussions, class activities, case studies.

The strongest frameworks will demonstrate evidence of reflection, synthesis, and intellectual development.

## Looking Ahead

### Week 14

**Future Challenges**

**Stress Testing Frameworks**

**Course Reflection**

**Course Evaluation**

Week 14 serves several purposes.

Students will: examine future challenges, stress-test their frameworks, reflect upon the semester, participate in course evaluation.

This slide helps students understand that the framework assignment is not the endpoint. It becomes the starting point for the final discussion.

Students should arrive next week prepared to defend, revise, and refine their frameworks.

CLOSING REFLECTION

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*What principle are you most unwilling to compromise?*

SEID 2364 • Week 13

33

The final question intentionally returns the focus to the individual.

"What principle are you most unwilling to compromise?"

This question is useful because it forces prioritization. Most students can generate a long list of ethical values. The challenge is identifying which principles remain most important when difficult tradeoffs arise.

Allow students time to reflect privately. Public discussion is optional.

The purpose is not consensus. The purpose is to leave students contemplating the core commitments that anchor their emerging ethical frameworks as they move toward the final assignment and the final week of the course.